School of Computer Science Engineering and Technology

BENNETT UNIVERSITY
TIMES OF INDIA GROUP

Course- B.Tech.	Type- Core
Course Code- **2025CSET106**	Course Name- **Discrete Mathematical Structures**
Year- I	Semester- Even (2^{nd} Semester)

Tutorial-2

Tutorial No.	**Name**	**CO1**	**CO 2**	**CO3**
2	Truth values of compound propositions using truth table and without truth table, and logical equivalences	✓	--	✓

Objective: The main objective of this tutorial is to learn about propositions, truth values of compound propositions using truth table and without truth table, and logical equivalences.

1. Given the truth values of p and q as true and those of r and s as false, find the truth values of the following:
   (a) $p \vee (q \wedge r)$
   (b) $(p \wedge (q \wedge r)) \vee \neg((p \vee q) \wedge (r \vee s))$

2. Construct the truth table for the following:
   (a) $[p \wedge (p \rightarrow q)] \rightarrow q$
   (b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
   (c) $(p \rightarrow q) \wedge (\neg p \rightarrow r)$
   (d) $(p \leftrightarrow q) \vee (\neg q \leftrightarrow r)$
   (e) $(\neg p \leftrightarrow \neg q) \leftrightarrow (q \leftrightarrow r)$

3. What is the value of x after each of these statements is encountered in a computer program, if x = 1 before the statement is reached?
   (a) if x + 2 = 3 then x := x + 1
   (b) if (x + 1 = 3) OR (2x + 2 = 3) then x := x + 1
   (c) if (2x + 3 = 5) AND (3x + 4 = 7) then x := x + 1
   (d) if (x + 1 = 2) XOR (x + 2 = 3) then x := x + 1
   (e) if x < 2 then x := x + 1

4. Use De Morgan's laws to find the negation of each of the following statements.
   (a) Amar is rich and happy.
   (b) Virat is smart and hard working.
   (c) Leela will move to Bangalore or Delhi.

5. Establish the following logical equivalences by developing a sequence of logical equivalences.
   Note: Do not use truth table.

(a) $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$

(b) $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$

(c) $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$

(d) $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$

(e) $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

(f) $\neg p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \vee r)$

(g) $p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$

(h) $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

"***Dream is not that which you see while sleeping it is something that does not let you sleep***."

A.P.J. Abdul Kalam